

# Homogeneous Systems

## Definition

**Homogeneous:** A system of equations is **homogeneous** if it can be written in the form  $A\mathbf{x} = \mathbf{0}$ .

## Example

$$\begin{aligned}x_1 + x_2 &= 1 \\x_1 - 4x_2 &= 0\end{aligned}$$

$$\begin{aligned}x_1 + x_2 &= 0 \\x_1 - 4x_2 &= 0\end{aligned}$$

Every homogeneous system has a **trivial solution** where  $\mathbf{x} = \mathbf{0}$ .

# Nontrivial Solutions Existence

Recall that the system  $A\mathbf{x} = \mathbf{b}$  for all  $\mathbf{b} \in \mathbb{R}^m$  has:

- ▶ A unique solution if  $A$  has a pivot in every row
- ▶ Infinite number of solutions if  $A$  has a solution and at least one free variable

## Theorem

*The homogeneous equation  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution if and only if the solution set has at least one free variable.*

## Solution Sets of Consistent Systems

Find the solution set to the homogeneous system write in vector form (from lecture 4):

$$x_1 + 3x_2 - 2x_3 = 0$$

$$-2x_1 - 6x_2 + 4x_3 = 0$$

## More Practice!

Find the solution set to the **nonhomogeneous** system

$$x_1 + 3x_2 + 7x_3 = 4$$

$$x_2 + 3x_3 = 5$$

$$-2x_1 - 4x_2 - 8x_3 = 2$$

# Comparing Homogeneous and Nonhomogeneous Solutions

$$\begin{aligned}x_1 + 3x_2 + 7x_3 &= 0 \\x_2 + 3x_3 &= 0 \\-2x_1 - 4x_2 - 8x_3 &= 0\end{aligned}$$

$$\begin{aligned}x_1 + 3x_2 + 7x_3 &= 4 \\x_2 + 3x_3 &= 5 \\-2x_1 - 4x_2 - 8x_3 &= 2\end{aligned}$$

# Infinite Solutions from Homogeneous System

## Theorem

Let  $A\mathbf{x} = \mathbf{b}$  be a consistent system with solution  $\mathbf{x} = \mathbf{p}$ . Then there exists a solution set of vectors of the form

$$\mathbf{x} = \mathbf{p} + t\mathbf{v}.$$

Where  $t\mathbf{v}$  is a solution to the *homogeneous* system  $A\mathbf{x} = \mathbf{0}$ .

## Proof.

We will show this by showing  $A\mathbf{x} = A(\mathbf{p} + t\mathbf{v}) = \mathbf{b}$ .

$$\begin{aligned} A\mathbf{x} &= A(\mathbf{p} + t\mathbf{v}) \\ &= A\mathbf{p} + A(t\mathbf{v}) \\ &= \mathbf{b} + \mathbf{0} \\ &= \mathbf{b} \end{aligned}$$

